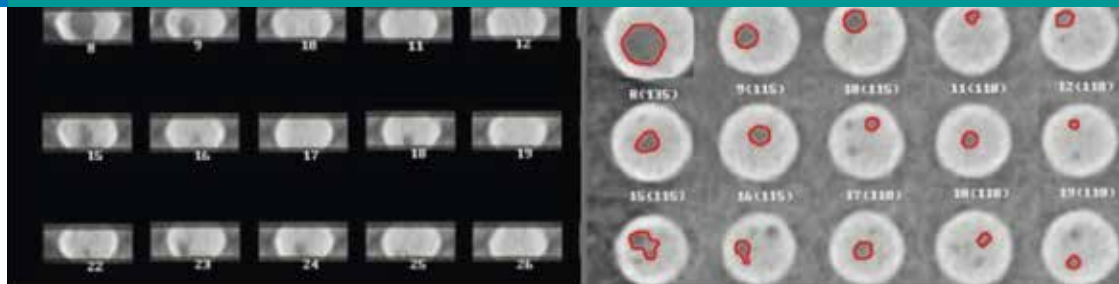


YXLON FF65 CL

World's best resolution 3D X-ray inspection system
for fully automated verification of IC packaging defects

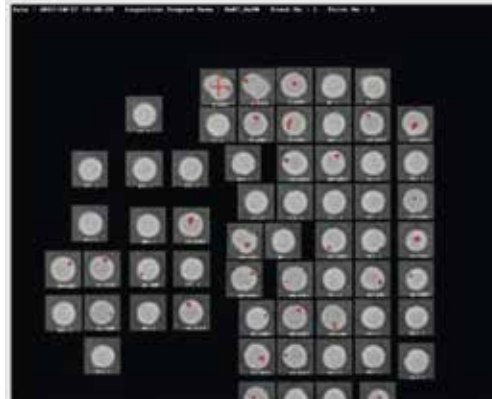
- High-speed 3D AXI brings about a revolution in the semiconductor process management
- Most precise measurements of defect dimensions for advanced 3D IC packaging, MEMS and Sensors
- Best reliable and reproducible check of process conditions and defect parameters
- Leading-edge simple and easy operation with fully automated wafer handling and testing process



Inspection result list:

Point No.	Pin No.	Type	Judge	Judgment table	X[μm]	Y[μm]	HeadBumpArea[μm ²]	ZGravityPosition[μm]	Top2BottomAlignment[μm]	BumpVolume[μm ³]
1	5	BGA	NG	C-Size-Test	1506	342	8226.740	124.000	▲5.100	0.000
1	2	BGA	NG	C-Size-Test	2024	344	9279.760	131.000	▲2.820	0.000
1	1	BGA	NG	C-Size-Test	815	345	8530.450	130.000	▲1.000	0.000
1	29	BGA	NG	C-Size-Test	974	387	8289.560	123.000	▲4.560	0.000
1	15	BGA	OK	C-Size-Test	747	685	9149.130	129.000	▲1.430	0.000
1	43	BGA	OK	C-Size-Test	220	1287	8899.830	130.000	0.660	0.000
1	24	BGA	OK	C-Size-Test	1183	824	7632.850	131.000	1.660	0.000
1	17	BGA	OK	C-Size-Test	1183	663	8642.560	130.000	1.390	0.000
1	49	BGA	OK	C-Size-Test	1246	1206	8494.980	133.000	1.370	0.000
1	13	BGA	OK	C-Size-Test	329	687	8523.890	137.000	2.060	0.000
1	16	BGA	OK	C-Size-Test	966	662	7889.590	134.000	1.740	0.000
1	24	BGA	OK	C-Size-Test	356	686	9306.680	129.000	1.330	0.000
1	12	BGA	OK	C-Size-Test	2683	499	8554.810	134.000	2.050	0.000
1	61	BGA	OK	C-Size-Test	1526	1646	9030.460	131.000	1.890	0.000
1	19	BGA	OK	C-Size-Test	1509	662	8339.420	133.000	2.480	0.000
1	45	BGA	OK	C-Size-Test	571	1284	7821.880	133.000	2.210	0.000

Insip Image



Judgment table C-Size-Test

Edit Judgment table

	HeadBumpArea[μm ²]	ZGravityPosition[μm]	Top2BottomAlignment[μm]	BumpVolume[μm ³]
Upper limit	1000000000.000	1000000000.000	2.750	1000000000.000
Lower limit	0.000	0.000	0.000	0.000

Statistical inspection result

	HeadBumpArea[μm ²]	ZGravityPosition[μm]	Top2BottomAlignment[μm]	BumpVolume[μm ³]
Max	9530.050	137.000	4.560	0.000
Min	7482.840	129.000	0.660	0.000
Avg.	4130.973	66.615	0.933	0.000
Std Dev.	4362.198	68.903	1.012	0.000
Total NG	0	0	3	0



YXLON
Technology with Passion

The exciting fully automated inspection system for 3D inspection and measurement

Semiconductor manufacturing requires automated, high quality, reliable, fast and non-destructive inspection and analysis for an optimum production as defects can be found on the wafer, on a substrate, on a strip or in the subassembly of a final device. The new X-ray inspection system YXLON FF65 CL has especially been designed to provide the best automatic analysis of smallest and most demanding features within the 3D IC, MEMS and sensors. The result: impressively precise and reproducible test and inspection excellence.

The FF65 CL uses 2D and 3D X-ray techniques that are based on Yxlon's renowned technology and innovation strengths: the open micro-focus X-ray sources with high-power targets, the latest generation of optimized high-

resolution X-ray detectors with long service life, and a high-precision, fully automated manipulation system. Collectively, these leading-edge components make the FF65 CL the perfect solution for the automatic analysis of bumps, flat solder connections, and filled vias to easily and reproducibly find non-wetted bumps, voiding and misalignments.

As the market leader for X-ray, Computed Laminography (CL), and Computed Tomography (CT) inspection systems in the electronics industry, Yxlon is continuously innovating and providing solutions precisely where they are needed. The YXLON FF65 CL pursues this tradition to ensure that the semiconductor industry is completely supported for today's and tomorrow's manufacturing challenges.

Our global, interlinked, service network with seven regional service centers and more than 50 local partners is prepared to quickly and efficiently support and help our customers in order to ensure highest quality and minimum downtimes.

Applications for the FF65 CL

Solder bump and filled via analysis for:

- Micro-Bumps/C4s, TSVs, Micro-Vias and Cu-Pillars in today's advanced packaging including:
- PoP, 3D IC, MEMS, Sensors, SiP, flip chip and their sub-assemblies

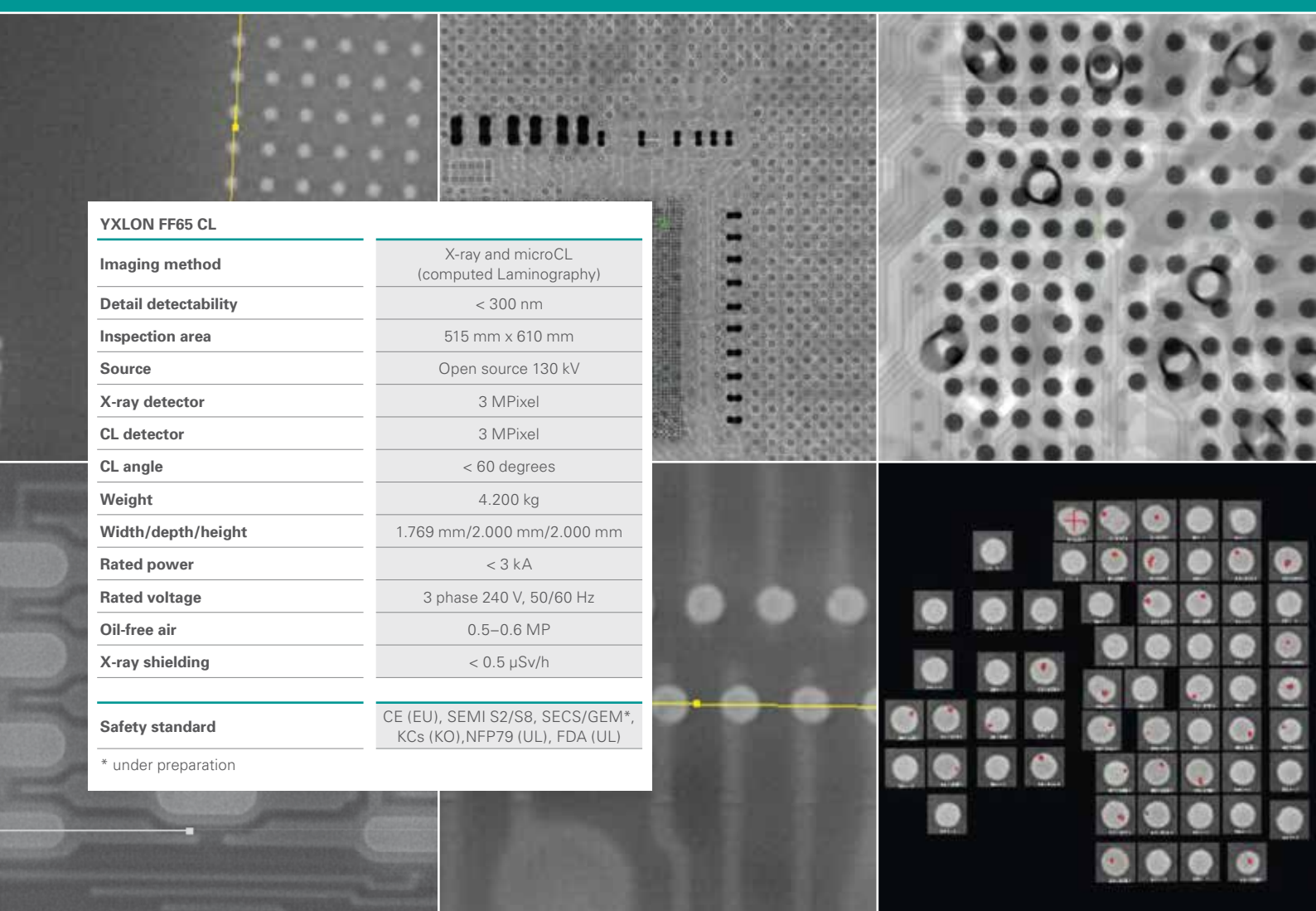
YXLON FF65 CL

Imaging method	X-ray and microCL (computed Laminography)
Detail detectability	< 300 nm
Inspection area	515 mm x 610 mm
Source	Open source 130 kV
X-ray detector	3 MPixel
CL detector	3 MPixel
CL angle	< 60 degrees
Weight	4.200 kg
Width/depth/height	1.769 mm/2.000 mm/2.000 mm
Rated power	< 3 kA
Rated voltage	3 phase 240 V, 50/60 Hz
Oil-free air	0.5–0.6 MP
X-ray shielding	< 0.5 µSv/h

Safety standard

CE (EU), SEMI S2/S8, SECS/GEM*, KCs (KO), NFP79 (UL), FDA (UL)

* under preparation





**Our experience is
the key to your success**



The perfect solution for high-volume, automated and reliable semiconductor joint analysis

The YXLON FF65 CL is distinguished by its large inspection area of 510 x 610 mm and the detectability of < 300 nm, making it ideal for automatically and non-destructively analyzing solder bumps and filled vias in 3D ICs, flip chips and wafers.

The innovative vacuum mechanism of the system manipulator holds the sample securely and precisely during analysis and counteracts the effects of sample warpage.

The FF65 CL provides 2D (top-down) with a high-performance flat-panel and 3D (CL – Computed Laminography) automated analysis using a high-resolution Image Intensifier within a special manipulation assembly for its inclined rotations. The latest generation of nano-

focus X-ray tubes creates 2D and 3D images that can reveal and measure the smallest voids and features. This enables the YXLON FF65 CL to analyze the most demanding advanced semiconductor challenges.

A user-friendly and intuitive graphical user interface (GUI) allows the easy creation of automated, multi-point and multi-functional analysis inspection programs.

Measurement repeatability over time is ensured by automatic, continuously monitoring background calibration tests over all aspects of the system.

System properties at a glance:

- Automated high-throughput analysis with the best reproducibility and reliability of the results
- Simple creation of automated, multi-point and multi-functional analysis inspection programs allowing rapid change between sample and measurement tasks
- Continuous background monitoring and optimization to ensure measurement repeatability and accuracy
- Fast and reproducible accuracy of all measurements

Your benefits for semiconductor production

- Improved quality monitoring – inspect more locations at better resolution to identify failures that otherwise could be missed
- Significant cost reduction through better test coverage leading to improved yield
- Reliable and reproducible check of consistency of process and defect parameters at any time
- This innovative automated analysis solution is easy to use, optimizing cost of operation

			
Feature	FF65 IL	FF65 CL	FF70 CL
Top Down Inspection Coverage	350 x 350 mm	400 x 296 mm [515 x 500 mm]*	515 x 610 mm
PCT Inspection Coverage	350 x 350 mm	400 x 292 mm [495 x 480 mm]*	475 x 570 mm
Max. Sample Size	350 x 350 mm	400 x 300 mm [515 x 610 mm]*	515 x 610 mm
Min. Sample Size	80 x 50 mm	30 x 30 mm [30 x 30 mm]*	30 x 30 mm
Sample Holder	Automatically adjustable conveyor	Standard jig [Carbon-fiber vacuum jig]*	Carbon-fiber vacuum jig
Warpage Allowance	3 mm	2 mm	Vacuum jig
Max. Sample Weight	2 kg	2 kg	2 kg
	*optional		

YXLON
Technology with Passion

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